

Shuangbiao Liu
e-mail: liusb@northwestern.edu
Department of Mechanical Engineering,
Northwestern University,
Evanston, IL 60208

Sylvie Lannou
Département de Mécanique,
Ecole Polytechnique,
F-91128 Palaiseau, France

Qian Wang

Leon Keer

Department of Mechanical Engineering,
Northwestern University,
Evanston, IL 60208

Solutions for Temperature Rise in Stationary/Moving Bodies Caused by Surface Heating With Surface Convection

Determining the temperature rise of contact interfaces subject to surface heating is essential to controlling thermally induced failures in manufacturing processes and tribology. This paper provides a summary of solutions to problems of a stationary/moving half-space or half-plane without/with surface convection. In the first two sections, basic formulations for bodies with negligible surface convection are grouped as explicitly and completely as possible in terms of the Green's function, the influence coefficients, and the frequency response function. In the final section, the influence coefficients are applied to solve surface-heating problems with surface convection. The time required to reach approximately steady state is found for bodies subject to a unit heat flux. The effect of convection is found to be dependent on the Péclet number and location. [DOI: 10.1115/1.1795234]

Introduction

Surface heating is a common phenomenon in manufacturing processes [1] and tribology [2]. For example, the frictional loss between two meshing gear teeth is converted into surface heat flux. More importantly, surface heating is one of the culprits of gear scuffing [3]. Modeling the temperature rise due to surface heating, especially the surface temperature rise known as the flash temperature [4], is essential to the failure study of cutting tools and contact components. Temperature effects on metallurgical microstructure, thermal shrinkage, thermal cracking, residual stresses, and chemical modifications of materials are collectively studied as surface integrity problems, which are significant to the performance and reliability analyses of mechanical components. The modeling of temperature rise could date back to the beginning of last century [5].

Surface-heating problems of a stationary or moving half-space and half-plane are complicated due to time dependency, component motion, heat source distribution [6,7], and boundary conditions. This paper summarizes known results, while exploring a more complete set of solutions to surface-heating problems without surface convection that are presented by Carslaw and Jaeger [5]. These results can be applied to develop solutions to problems with surface convection. In the first two sections, basic formulations are grouped into three categories, with an adiabatic boundary in the nonheating surface: Green's functions (Beck et al. [8] discussed the Green's functions in great depth), influence coefficients, and frequency response functions in order to facilitate analytical and numerical modeling. These formulations are applied to solve surface-heating problems with a surface convection boundary in the final section. Three cases are discussed according to the time variation of the heat source: a) an instantaneous case (TI) with a varying heat source, where "T" indicates the transient characteristics; b) a continuous case (TC) with a constant heat source; and c) a steady state case (SS), which is a special TC case with infinite time. Table 1 summarizes the available fundamental formulas in a closed form, i.e., the formulas are free of integrals but may contain special mathematical functions such as the error function, the complementary error function, the exponential inte-

gral, and the Bessel functions. Two new items presented in this paper are identified by "#" in Table 1. Note that in all cases, the Péclet numbers are not a function of spatial coordinates. In addition, the motion is time independent in the TC and SS cases, while it could be a function of time in the TI cases.

The time required to reach approximately steady state is derived for bodies subject to a unit heat flux. For a half-plane, a reference point [9] is required for TC and SS cases. It is found that a half-plane requires less time than a half-space to achieve steady state, where a reference point is located one unit away from the origin. If the surface heat-transfer coefficient is not negligible, results in Section 3 show that under a small Péclet number (less than three), all temperature values are reduced; while under larger Péclet numbers, the temperature values are significantly reduced at locations on the leading side of the surface-heating region.

1 Surface Heating of a Half-Space

An irregularly distributed heat source $\bar{q}$ per unit time per unit area is applied on the surface of a half-space [Fig. 1(a)] with uniform initial temperature. A nondimensional coordinate system, $x_j = \bar{x}_j/l$, where l is a characteristic length, is fixed to the heat source. All barred variables are dimensional. The half-space is stationary or moving relative to the heat source (and the coordinate system) with speeds, $\bar{V}_1$ and $\bar{V}_2$ ($\bar{V}_3 = 0$), which can vary with time, but not with position. One could combine two speeds into a single speed and adjust the x_1 coordinate to be in the speed direction. Material properties of the half-space are diffusivity κ and thermal conductivity K . The nondimensional heat source ($q = \bar{q}/q_0$) will cause a nondimensional temperature rise, $T = TK/(lq_0)$, governed by the heat-conduction equation,

$$T_{,ii} = \frac{\partial T}{\partial t} + \text{Pe}_1 T_{,1} + \text{Pe}_2 T_{,2} \quad (1.1)$$

where t is nondimensional time, $t = \kappa \bar{t}/l^2$; Pe_j is the Péclet number in the x_j direction, $\text{Pe}_j = \bar{V}_j l / \kappa$. In the surface-heated region S , a boundary condition is specified as

$$-T_{,3} = q \quad (1.2)$$

Outside S , an adiabatic boundary condition is applied. Alternatively, a convection boundary condition could be applied (see Sec. 3).

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Table 1 Equation number of fundamental solutions in a closed form

Body	Case	Green's Functions		Influence Coefficients		Frequency Response
		Stationary	Moving	Stationary	Moving	Functions
Half-space	TI	1.4	1.5		1.17	1.24a
	TC	1.10	1.7 [#]		NA	1.24b
Half-plane	SS		1.12	1.21a [#]	Large Pe [11]	1.24c
	TI		2.3		2.9	2.21a
	TC	2.4	NA	2.11a in the surface	NA	2.21b
	SS	2.6	2.5	2.20	2.13 in the surface	2.21c

1.1. Green's Functions

1.1.1 Instantaneous Case. One can find the following solution with the Green's function [5] for the *stationary* half-space:

$$T(x_1, x_2, x_3, t) = \int_0^t \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} G^{TI}(x_1 - x'_1, x_2 - x'_2, x_3, t - t') q(x'_1, x'_2, t') dx'_1 dx'_2 dt' \quad (1.3)$$

$$G^{TI}(x_1, x_2, x_3, t) = \frac{\exp[-R^2/(4t)]}{4(\pi t)^{3/2}} \quad (1.4)$$

where $R^2 = x_1^2 + x_2^2 + x_3^2$. Note that in the denominator, four is used rather than eight ([5], p. 256) because half-space, not infinite space, problems are studied. The Green's function in Eq. (1.4) can be extended to the *moving* half-space with time-dependent Péclet numbers,

$$G^{TI}(x_1, x_2, x_3, \Delta t) = \frac{\exp[-R'^2/(4\Delta t)]}{4(\pi\Delta t)^{3/2}} \quad (1.5)$$

where $\Delta t = t - t'$ and $R'^2 = (x_1 - \int_{t'}^t Pe_1 d\tau)^2 + (x_2 - \int_{t'}^t Pe_2 d\tau)^2 + x_3^2$. The general expression of the temperature rise consists of four integrals: one in R' and the remaining three in Eq. (1.3), which are costly for numerical computation and require analytical simplification if possible.

1.1.2 Continuous Case. When the heat source is constantly applied from time 0 to t and the half-space has only a time-invariant Péclet number in the x_1 direction ($Pe_2 = 0$), then the temperature rise at time t is

$$T(x_1, x_2, x_3, t) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} G^{TC}(x_1 - x'_1, x_2 - x'_2, x_3, t) q(x'_1, x'_2) dx'_1 dx'_2 \quad (1.6)$$

$$\begin{aligned} G^{TC}(x_1, x_2, x_3, t) &= \int_0^t G^{TI}(x_1, x_2, x_3, t - t') dt' \\ &= \frac{1}{4\pi R} \exp\left[\frac{-Pe_1(R - x_1)}{2}\right] \left[2 - \operatorname{erfc}\left(\frac{Pe_1\sqrt{t}}{2} - \frac{R}{2\sqrt{t}}\right) \right. \\ &\quad \left. + \exp(RPe_1) \operatorname{erfc}\left(\frac{Pe_1\sqrt{t}}{2} + \frac{R}{2\sqrt{t}}\right) \right] \end{aligned} \quad (1.7)$$

where the integration with respect to time has been carried out. The above Green's function is probably novel since reported results are not explicitly integrated. For example, Hou and Komanduri [7] recently presented the following integral, rewritten in current nondimensional form with the speed pointing in the positive x_1 direction, as

$$\begin{aligned} G^{TC}(x_1, x_2, x_3, t) &= \frac{Pe_1}{8\pi^{3/2}} \exp\left(\frac{Pe_1 x_1}{2}\right) \int_0^{Pe_1^2 t/4} \\ &\quad \times \left(-\xi - \frac{R^2 Pe_1^2}{16\xi} \right) \frac{d\xi}{\xi^{3/2}} \end{aligned} \quad (1.8)$$

Equation (1.7) can be obtained from Eq. (1.8) by using the following integral

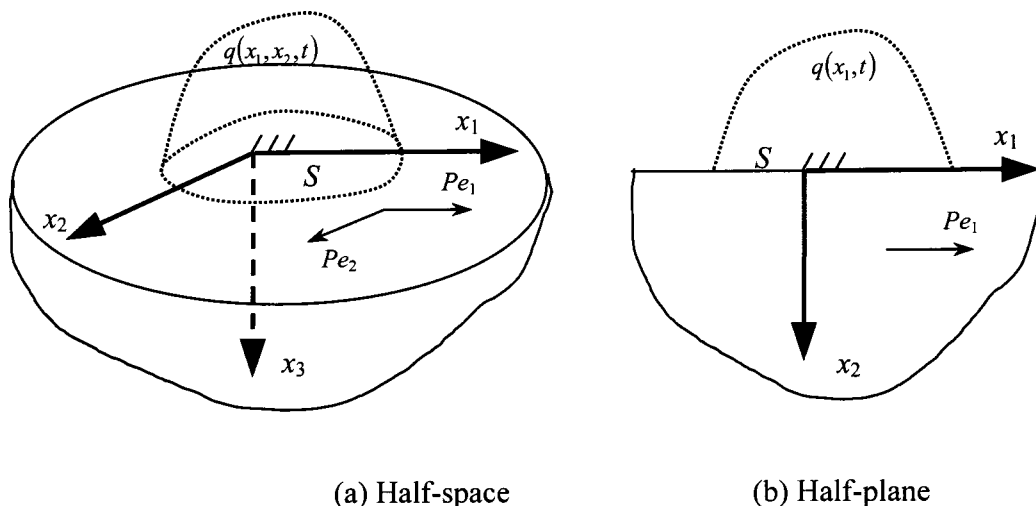


Fig. 1 Bodies subject to the surface heat source

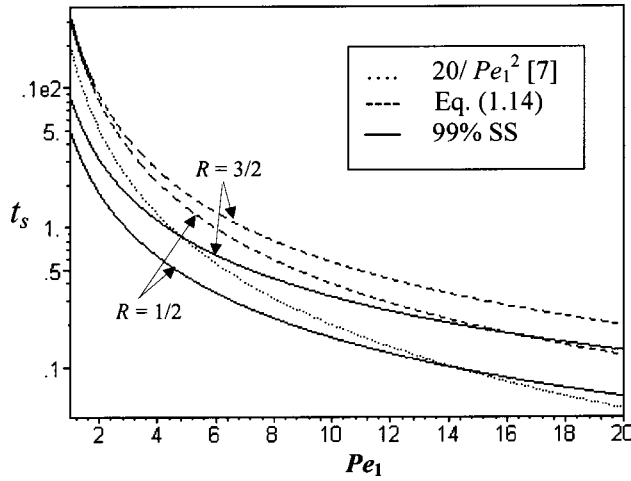


Fig. 2 Effect of the Péclet number on the time required to reach approximately steady state

$$\begin{aligned} & \int \exp\left(-C^2\xi - \frac{D^2}{\xi}\right) \frac{d\xi}{\xi^{3/2}} \\ &= -\sqrt{\pi}[\exp(-2CD)\operatorname{erfc}(C\sqrt{\xi} - D/\sqrt{\xi}) \\ & \quad - \exp(2CD)\operatorname{erfc}(D/\sqrt{\xi} + C\sqrt{\xi})]/(2D) \end{aligned} \quad (1.9)$$

where C and D are independent of ξ .

For the *stationary* half-space, Eq. (1.7) becomes the known solution (p. 261 [5]),

$$G^{TC}(x_1, x_2, x_3, t) = \frac{1}{2\pi R} \operatorname{erfc}\left(\frac{R}{2\sqrt{t}}\right), \quad \text{Pe}_1 = 0 \quad (1.10)$$

1.1.3 Steady-State Case. The SS solution for the half-space can be found simply by letting time approach infinity in Eq. (1.7) (alternatively, see p. 267 [5])

$$\begin{aligned} T(x_1, x_2, x_3) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} G^{SS}(x_1 - x'_1, x_2 - x'_2, x_3) \\ & \quad \times q(x'_1, x'_2) dx'_1 dx'_2 \end{aligned} \quad (1.11)$$

$$G^{SS}(x_1, x_2, x_3) = \frac{1}{2\pi R} \exp\left[\frac{-\text{Pe}_1(R - x_1)}{2}\right] \quad (1.12)$$

It is interesting to determine time t_s when the steady state is approximately reached. Hou and Komanduri argued that the contribution to the integration beyond five is negligible in Eq. (1.8), so that the upper limit of the integration could be set approximately as five. Therefore, it was concluded in their work that the steady state could have been established at approximately $t_s = 20/\text{Pe}_1^2$, which corresponds to dimensional time of $20\kappa/V^2$. From Eqs. (1.7)–(8), t_s should be a function of the Péclet number (Pe_1) and the radius (R). The ratio of the TC result to the SS result gives a quantitative criterion to evaluate the degree to which the steady state is reached. Given this ratio, the location, and the Péclet number, one can accurately find t_s by substituting Eq. (1.7) into the ratio expression. When this ratio is set as 0.99 at $R=0.5$ or 1.5, the corresponding time is depicted in Fig. 2 (solid line) with respect to the Péclet number. Figure 2 shows that the Hou and Komanduri result (dotted line) overestimates the time for low Péclet number and underestimates the time for high Péclet number.

The Green's function of TC cases involves the complementary error function, which decays to zero faster than the exponential function. It is reasonable to claim that the steady state is approximately reached when the following condition is fulfilled,

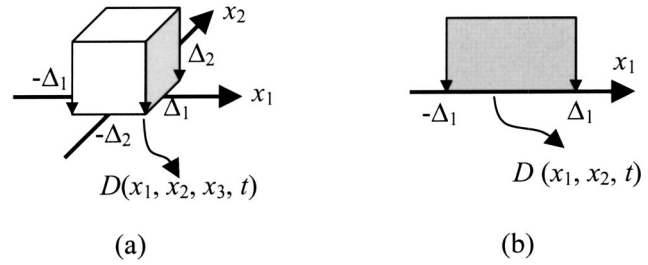


Fig. 3 Schematics of influence coefficients

$$\frac{\text{Pe}_1 \sqrt{t_s}}{2} - \frac{R}{2\sqrt{t_s}} = n \quad (1.13)$$

where n is 3 or larger since $\operatorname{erfc}(3)$ is $O(10^{-5})$. After rewriting, one can find t_s

$$t_s = [(n + \sqrt{n^2 + R\text{Pe}_1})/\text{Pe}_1]^2 \quad (1.14)$$

which has a square root term and thus differs from Hou and Komanduri [7] ($t_s \sim \text{Pe}_1^{-2}$). Figure 2 also depicts results obtained from Eq. (1.14) with $n=3$, where the curve trends reveal that the smaller the Péclet number, the longer the time to reach the steady state. From Eq. (1.14), it is seen that locations far from the origin require a longer time to reach the steady state.

Equation (1.14) is not applicable for zero Péclet number (the *stationary* half-space); instead the term, $\operatorname{erfc}[R/(2\sqrt{t_s})]$, in Eq. (1.10) should go toward one as time increases. If the ratio of the TC result to the SS result is denoted as $1 - e$, then t_s can be found as follows:

$$t_s \approx R^2/(e^2 \pi) \quad (1.15)$$

For example, set e to 0.01 so that the ratio is 0.99. If the half-space is *moving* with $\text{Pe}_1 = 1$, t_s is approximately 5 at $R = \frac{1}{2}$. However, t_s should be around 800 for the same location in the *stationary* half-space. From the comparison between Eq. (1.14) and Eq. (1.15), it is seen that the location has greater effect on t_s in *stationary* bodies than in *moving* ones. It should be pointed out that these equations for t_s are valid for locations inside the half-space as well. However, since the equations for t_s are obtained from Green's functions, they are only accurate in the case of a unit heat source. For problems with arbitrary heat sources, these equations are applicable to locations far away from the heat source.

1.2 Influence Coefficients. Influence coefficients are responses to a constant heat distribution on the surface over the rectangular patch $x_1 \in [-\Delta_1, \Delta_1]$ and $x_2 \in [-\Delta_2, \Delta_2]$ [Fig. 3(a)]. Since an arbitrary heat source can be discretized into a set of uniformly distributed heat sources, influence coefficients are widely used in numerical simulation. The influence coefficients for a half-space are generally written as the following double integrals:

$$D(x_1, x_2, x_3, t) = \int_{-\Delta_1}^{\Delta_1} \int_{-\Delta_2}^{\Delta_2} G(x_1 - x'_1, x_2 - x'_2, x_3, t) dx'_1 dx'_2 \quad (1.16)$$

For the steady state, the variable t in Eq. (1.16) should be absent.

1.2.1 Instantaneous Case. Integrating the Green's function for the *stationary* half-space leads to

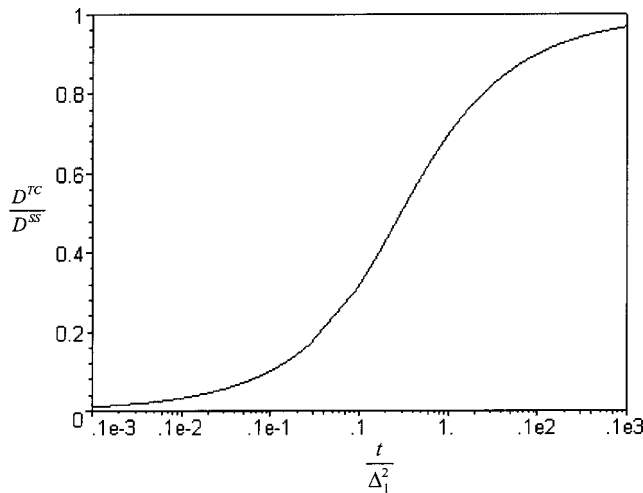


Fig. 4 Evolution of the influence coefficient at the origin with time

$$D^{TI}(x_1, x_2, x_3, t) = \frac{\exp[-x_3^2/(4t)]}{4\sqrt{\pi t}} \left[\operatorname{erf}\left(\frac{x_1 + \Delta_1}{2\sqrt{t}}\right) - \operatorname{erf}\left(\frac{x_1 - \Delta_1}{2\sqrt{t}}\right) \right] \left[\operatorname{erf}\left(\frac{x_2 + \Delta_2}{2\sqrt{t}}\right) - \operatorname{erf}\left(\frac{x_2 - \Delta_2}{2\sqrt{t}}\right) \right] \quad (1.17)$$

For the *moving* half-space, t and x_i in the above equation should be replaced by $t - t'$ and $x_i - \int_{t'}^t \text{Pe}_i d\tau$, respectively.

1.2.2 Continuous Case. An attempt by the authors to express the influence coefficients in a closed form was unsuccessful. However, instead of double integrals (Eq. 1.16) the influence coefficients for the *stationary* half-space could be written as a single integral with respect to time based on Eq. (1.17) as follows:

$$D^{TC}(x_1, x_2, x_3, t) = \int_0^{\sqrt{t}} \frac{\exp[-x_3^2/(4\tau^2)]}{2\sqrt{\pi}} \left[\operatorname{erf}\left(\frac{x_1 + \Delta_1}{2\tau}\right) - \operatorname{erf}\left(\frac{x_1 - \Delta_1}{2\tau}\right) \right] \left[\operatorname{erf}\left(\frac{x_2 + \Delta_2}{2\tau}\right) - \operatorname{erf}\left(\frac{x_2 - \Delta_2}{2\tau}\right) \right] d\tau \quad (1.18a)$$

where $\tau = \sqrt{t - t'}$ is used. The maximum value is at the origin and can be obtained numerically by evaluating

$$D^{TC}(0,0,0,t) = \frac{2}{\sqrt{\pi}} \int_0^{\sqrt{t}} \operatorname{erf}\left(\frac{\Delta_1}{2\tau}\right) \operatorname{erf}\left(\frac{\Delta_2}{2\tau}\right) d\tau \quad (1.18b)$$

For the *moving* half-space, x_i in Eq. (1.18a) should be replaced by $x_i - \int_{t-\tau}^t \text{Pe}_i du$. From the computational point of view, a single integration can be handled more efficiently than double integration. Figure 4 shows the evolution of the influence coefficients with $\Delta_1 = \Delta_2$ at the origin normalized by the SS result of Eq. (1.21b), which is discussed in Sec. 1.2.3. Since the horizontal axis in Fig. 4 is time divided by Δ_1^2 , the value of t_s at the origin strongly depends on the size of the heated region and is roughly 100 to 1000 times Δ_1^2 .

1.2.3 Steady-State Case. For the *stationary* half-space, the Green's function has a kernel of the Newtonian potential, $G^{SS}(x_1, x_2, x_3) = 1/2\pi R$. From [10], the following integration can be found,

$$F(x_1, x_2, x_3) = \int \int \frac{1}{R} dx_1 dx_2 = \frac{x_1}{2} \ln\left(\frac{R+x_2}{R-x_2}\right) + \frac{x_2}{2} \ln\left(\frac{R+x_1}{R-x_1}\right) - x_3 \arctan\left(\frac{x_1 x_2}{x_3 R}\right) \quad (1.19)$$

$$\text{and } F(x_1, x_2, 0) = x_1 \ln(\sqrt{x_1^2 + x_2^2} + x_2) + x_2 \ln(\sqrt{x_1^2 + x_2^2} + x_1) \quad (1.20)$$

Therefore, the influence coefficients with $\text{Pe}_1 = 0$ can be expressed as,

$$D^{SS}(x_1, x_2, x_3) = [F(x_1 + \Delta_1, x_2 + \Delta_2, x_3) - F(x_1 - \Delta_1, x_2 + \Delta_2, x_3) - F(x_1 + \Delta_1, x_2 - \Delta_2, x_3) + F(x_1 - \Delta_1, x_2 - \Delta_2, x_3)] / (2\pi) \quad (1.21a)$$

The maximum value is at the origin and expressed as

$$D^{SS}(0,0,0) = \frac{2}{\pi} \left[\Delta_1 \ln \frac{\sqrt{s^2 + 1} + 1}{s} + \Delta_2 \ln(\sqrt{s^2 + 1} + s) \right] \quad (1.21b)$$

where $s = \Delta_1 / \Delta_2$ is the ratio of the heated region. If the patch is square (i.e., $s = 1$), $D^{SS}(0,0,0) = (4/\pi) \Delta_1 \ln(1 + \sqrt{2})$, which is approximately $2\Delta_1 / \sqrt{\pi}$ as obtained in [9].

When the half-space is *moving*, the influence coefficients may not be expressed in a closed form and numerical integration is used to obtain numerical values. However, Tichy [11] derived the temperature rise in the fast *moving* half-space under a uniform rectangular heat flux, which is, in fact, the influence coefficients for a very large Péclet number.

1.3 Frequency Response Functions. The general form of temperature rise in a mixed time and frequency domain can be expressed as [12],

$$\tilde{\tilde{T}}(\omega_1, \omega_2, x_3, t) = \int_0^t \frac{\tilde{q}(\omega_1, \omega_2, t')}{\sqrt{\pi \Delta t} \exp[\Delta t \omega^2 + x_3^2/(4\Delta t)]} dt' \quad (1.22)$$

where $\Delta t = t - t'$, $\omega^2 = \omega_1^2 + \omega_2^2 + i(\omega_1 \int_{t'}^t \text{Pe}_1 d\tau + \omega_2 \int_{t'}^t \text{Pe}_2 d\tau) / \Delta t$, and a double tilde ($\tilde{\tilde{\cdot}}$) denotes a two-dimensional Fourier transform with respect to x_1 and x_2 . The mixed domain involves the frequency domain with respect to x_1 and x_2 , the space domain of x_3 and the time domain.

If the Péclet number varies with time, numerical integration must be used to evaluate the time integrals. However, temperature in Eq. (1.22) is simplified below for problems with time-invariant Péclet numbers with the notation

$$w = \sqrt{\omega_1^2 + \omega_2^2 + i\omega_1 \text{Pe}_1 + i\omega_2 \text{Pe}_2} \quad (1.23)$$

For the *instantaneous* case, the integral in Eq. (1.22) is a convolution and the Fourier transform with respect to time changes the convolution to multiplication,

$$\frac{\tilde{\tilde{T}}(\omega_1, \omega_2, x_3, \omega_t)}{\tilde{q}(\omega_1, \omega_2, \omega_t)} = \frac{\exp(-x_3 \sqrt{w^2 + i\omega_t})}{\sqrt{w^2 + i\omega_t}} \quad (1.24a)$$

where ω_t is the frequency domain counterpart of time. A triple tilde ($\tilde{\tilde{\tilde{\cdot}}}$) denotes a three-dimensional Fourier transform with respect to x_1 and x_2 and time. For the *continuous* case and *steady-state* case, one can find the frequency response functions as follows, respectively:

$$\frac{\tilde{T}(\omega_1, \omega_2, x_3, t)}{\tilde{q}(\omega_1, \omega_2)} = \frac{1}{2w} \left[\exp(-x_3 w) \operatorname{erfc}\left(\frac{x_3}{2\sqrt{t}} - w\sqrt{t}\right) - \exp(x_3 w) \operatorname{erfc}\left(\frac{x_3}{2\sqrt{t}} + w\sqrt{t}\right) \right] \quad (1.24b)$$

$$\frac{\tilde{T}(\omega_1, \omega_2, x_3)}{\tilde{q}(\omega_1, \omega_2)} = \frac{\exp(-x_3 w)}{w} \quad (1.24c)$$

2 Surface Heating of a Half-Plane

An irregularly distributed heat source per unit time per unit length is applied on the surface of a half-plane [Fig. 1(b)] with uniform initial temperature. All quantities in this section are identical to those in Sec. 1 except that the x_2 coordinate is directed into the half-plane and the x_3 coordinate is absent. The temperature rise is governed by

$$T_{,ii} = \frac{\partial T}{\partial t} + \operatorname{Pe}_1 T_{,1} \quad (2.1)$$

In the surface-heated region S , a boundary condition is specified as

$$-T_{,2} = q \quad (2.2)$$

Outside S , an adiabatic boundary condition is applied. Solutions involving a convection boundary condition are presented in Sec. 3.

2.1 Green's Functions. The Green's functions for the half-plane can be found by integration of the half-space result. Carslaw and Jaeger [5] presented these Green's functions for the following cases:

- a. *Instantaneous case* with $\operatorname{Pe}_1 = 0$:

$$G^T(x_1, x_2, t) = \frac{1}{2\pi t} \exp\left(-\frac{R^2}{4t}\right) \quad (2.3)$$

where $R = \sqrt{x_1^2 + x_2^2}$. If the half-space is *moving*, t and x_1 in Eq. (2.3) should be replaced by $t - t'$ and $x_1 - \int_{t'}^t \operatorname{Pe}_1 d\tau$, respectively.

- b. *Continuous case* with $\operatorname{Pe}_1 = 0$:

$$G^{TC}(x_1, x_2, t) = \int_0^t G^T(x_1, x_2, t - t') dt' = \frac{-1}{2\pi} \operatorname{Ei}\left(-\frac{R^2}{4t}\right) \quad (2.4)$$

The Green's function of the *moving* half-plane for continuous cases is not available in a closed form.

- c. *Steady-state case*:

$$G^{SS}(x_1, x_2) = \frac{1}{\pi} \exp\left[\frac{\operatorname{Pe}_1 x_1}{2}\right] K_0\left[\frac{R \operatorname{Pe}_1}{2}\right] \quad (2.5)$$

The Green's functions in Eq. (2.4) for TC cases and Eq. (2.5) for SS cases are singular at the origin. Furthermore, Eq. (2.5) is everywhere infinity. The asymptotic behavior of modified Bessel functions can be shown to be $\lim_{x \rightarrow 0} K_0(cx) = -\ln c - \ln x + \ln 2 - \gamma$, where γ is the Euler's constant (0.5772). By ignoring a bulk value (the logarithmic infinity and constant terms), the Green's function with an appropriate reference point for the *stationary* half-plane is

$$G^{SS}(x_1, x_2) = -\frac{1}{\pi} \ln R \quad (2.6)$$

Recall that t_s is the time when the steady state is approximately

reached and the exponential integral for any small x can be written as $\operatorname{Ei}(-x) = \gamma + \ln x - x + O(x^2)$. If the ratio between the TC and SS results is given as $1 - e$, one can approximate t_s for the *stationary* half-plane by the following expression with the reference point at R_0 ,

$$t_s = \frac{R^2 - R_0^2}{8e(\ln R - \ln R_0)} \quad (2.7)$$

which is dependent on R_0 . The relationship between t_s and R here is different from that of the *stationary* half-space in Eq. (1.15). t_s is about 13.5 at $R = \frac{1}{2}$ with $e = 0.01$ and $R_0 = 1$. For the *moving* half-plane, t_s could be found numerically, and is about 3.1 at $R = \frac{1}{2}$ with $\operatorname{Pe}_1 = 1$, $e = 0.01$, and $R_0 = 1$. For this specific R_0 , the t_s of the half-plane is less than that of the half-space so in the neighborhood of $R = 1$, the half-plane reaches the steady state more rapidly than the half-space.

2.2 Influence Coefficients. Similar to half-space problems, the influence coefficients for the half-plane can be written as

$$D(x_1, x_2, t) = \int_{-\Delta_1}^{\Delta_1} G(x_1 - x'_1, x_2, t) dx'_1 \quad (2.8)$$

with a uniformly distributed heat source over $x_1 \in [-\Delta_1, \Delta_1]$ [Fig. 3(b)]. For the steady-state case, time should not be included in this expression.

2.2.1 Instantaneous Case. By integrating the Green's function of Eq. (2.3), one can obtain,

$$D^T(x_1, x_2, t) = \frac{1}{2\sqrt{\pi t}} \exp[-x_2^2/(4t)] \left[\operatorname{erf}\left(\frac{x_1 + \Delta_1}{2\sqrt{t}}\right) - \operatorname{erf}\left(\frac{x_1 - \Delta_1}{2\sqrt{t}}\right) \right] \quad (2.9)$$

If the half-plane is *moving*, t and x_1 in Eq. (2.9) should be replaced by $t - t'$ and $x_1 - \int_{t'}^t \operatorname{Pe}_1 d\tau$, respectively.

2.2.2 Continuous Case. It is convenient to write the influence coefficients in a time-integral form as following,

$$D^{TC}(x_1, x_2, t) = \frac{1}{\sqrt{\pi}} \int_0^{\sqrt{t}} \exp[-x_2^2/(4\tau^2)] \left[\operatorname{erf}\left(\frac{x_1 + \Delta_1}{2\tau}\right) - \operatorname{erf}\left(\frac{x_1 - \Delta_1}{2\tau}\right) \right] d\tau \quad (2.10)$$

If the half-plane is *moving*, x_1 in Eq. (2.10) should be replaced by $x_1 - \int_{t-\tau^2}^t \operatorname{Pe}_1 du$. In the surface of the *stationary* half-plane, Eq. (2.10) can be simplified due to the indefinite integral, $\int \operatorname{erf}(c/\tau) d\tau = \tau \operatorname{erf}(c/\tau) - c \operatorname{Ei}(-c^2/\tau^2)/\sqrt{\pi}$,

$$D^{TC}(x_1, 0, t) = \sqrt{\frac{t}{\pi}} \left[\operatorname{erf}\left(\frac{x_1 + \Delta_1}{2\sqrt{t}}\right) - \operatorname{erf}\left(\frac{x_1 - \Delta_1}{2\sqrt{t}}\right) \right] - \frac{x_1 + \Delta_1}{2\pi} \operatorname{Ei}\left[-\frac{(x_1 + \Delta_1)^2}{4t}\right] + \frac{x_1 - \Delta_1}{2\pi} \operatorname{Ei}\left[-\frac{(x_1 - \Delta_1)^2}{4t}\right] \quad (2.11a)$$

(see also p. 264 [5]). The maximum value is at the origin and can be expressed as

$$D^{TC}(0, 0, t) = 2\sqrt{\frac{t}{\pi}} \operatorname{erf}\left(\frac{\Delta_1}{2\sqrt{t}}\right) - \frac{\Delta_1}{\pi} \operatorname{Ei}\left(-\frac{\Delta_1^2}{4t}\right) \quad (2.11b)$$

2.2.3 Steady-State Case

2.2.3.1 *Surface temperature rise.* In the surface, one can calculate the influence coefficients by using the following integration (p. 269 [5]),

$$\int_0^a \exp(\pm u) K_0(u) du = a \exp(\pm a) \{K_0(a) \pm K_1(a)\} \mp 1 \quad (2.12)$$

where K_1 is the modified Bessel function of the second kind of order one. The influence coefficients can be written as

$$D^{SS}(x_1, 0) = \frac{1}{\pi} \begin{cases} \frac{(x_1 + \Delta_1) \exp(P_+) [K_0(-P_+) - K_1(-P_+)]}{(x_1 + \Delta_1) \exp(P_+) [K_0(P_+) + K_1(P_+)]}, & x_1 \leq -\Delta_1 \\ + (\Delta_1 - x_1) \exp(P_-) [K_0(-P_-) - K_1(-P_-)], & -\Delta_1 \leq x_1 \leq \Delta_1 \\ \frac{-(x_1 - \Delta_1) \exp(P_-) [K_0(P_-) + K_1(P_-)]}{(x_1 + \Delta_1) \exp(P_+) [K_0(P_+) + K_1(P_+)]}, & \Delta_1 \leq x_1 \end{cases} \quad (2.13)$$

where $P_{\pm} = \text{Pe}_1(x_1 \pm \Delta_1)/2$. The location of the maximum value is found between 0 and Γ_1 and is governed by

$$\exp(P_+) K_0(P_+) - \exp(P_-) K_0(P_-) = 0 \quad (2.14)$$

The numerical solution of x_1 for Eq. (2.14) with $\Gamma_1 = 1$ is plotted against the Péclet number in Fig. 5. The location of the maximum temperature apparently approaches the leading side of the heat source when the Péclet number increases. The maximum value can be determined by the middle expression in Eq. (2.13) with the numerical value of x_1 in Fig. 5. In Fig. 6, the influence coefficients with $\Gamma_1 = 1$ are shown for different Péclet numbers with Eq. (2.13) (solid lines without symbols). For a large Péclet number and thus a large $P_{\pm}$, it is convenient to use the following asymptotic behavior of the modified Bessel functions,

$$K_n(u) \approx \sqrt{\frac{\pi}{2u}} \exp(-u) \quad (2.15)$$

where u is large and n is an integer. By using this approximation, one can find the influence coefficients for large Pe_1

$$D^{SS}(x_1, 0) = \frac{2}{\sqrt{\pi \text{Pe}_1}} \begin{cases} 0, & x_1 < -\Delta_1 \\ \sqrt{x_1 + \Delta_1}, & -\Delta_1 \leq x_1 \leq \Delta_1 \\ \sqrt{x_1 + \Delta_1} - \sqrt{x_1 - \Delta_1}, & \Delta_1 < x_1 \end{cases} \quad (2.16)$$

which is identical to the results obtained by Ling et al. [9]. At the origin, it is obvious that $D^{SS}(0, 0) = 2\sqrt{\Delta_1}/(\text{Pe}_1\pi)$. The maximum value is at the leading edge of the heated region and equals $2\sqrt{2\Delta_1}/(\text{Pe}_1\pi)$.

For the *stationary* half-plane, the influence coefficients in Eq. (2.13) are infinite; however, these can be expressed in relative to the value at the origin, as

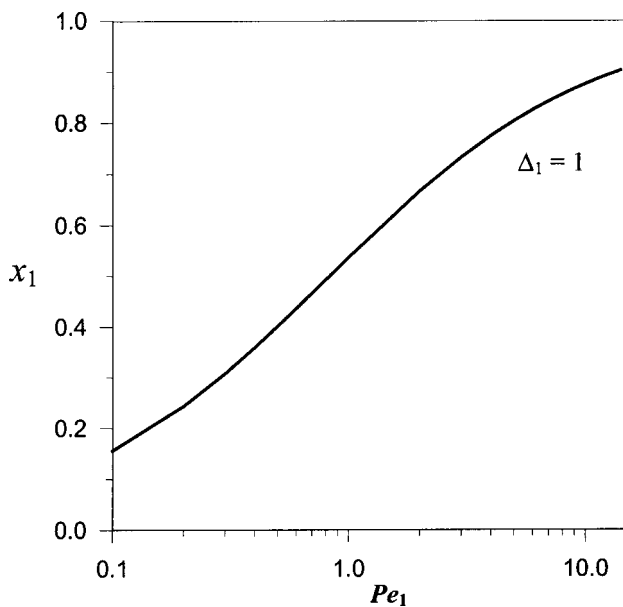


Fig. 5 Location of maximum temperature with the Péclet number

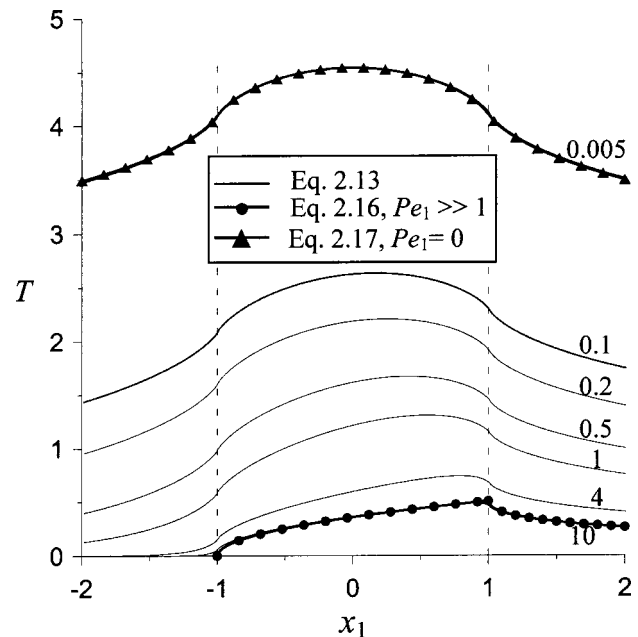


Fig. 6 SS surface temperature of the half-plane for various Péclet numbers. Those values around curves are Péclet number for Eq. (2.13).

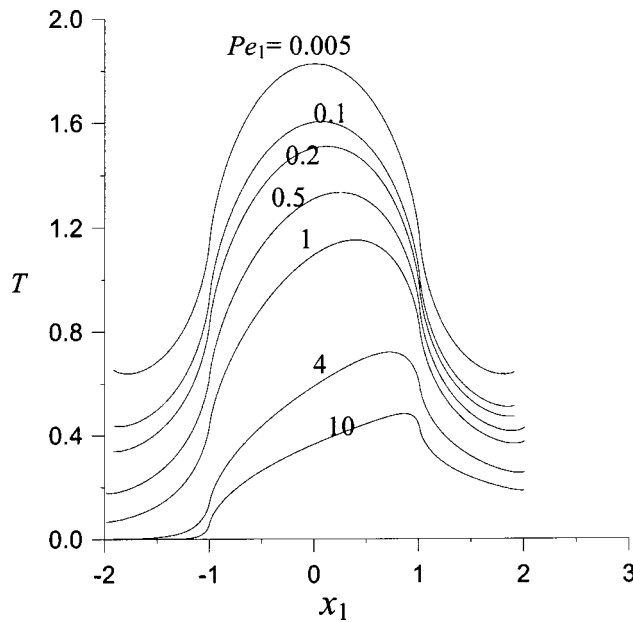


Fig. 7 SS surface temperature of the half-plane for different Peclet numbers ($h=1$)

$$D^{SS}(x_1, 0) - D^{SS}(0, 0) = \frac{1}{\pi} \left[\ln \left(\frac{|x_1 - \Delta_1|^{x_1 - \Delta_1}}{|x_1 + \Delta_1|^{x_1 + \Delta_1}} \right) + 2\Delta_1 \ln \Delta_1 \right], \quad (Pe_1 = 0) \quad (2.17)$$

where the asymptotic behaviors of the modified Bessel functions $\lim_{x \rightarrow 0} K_0(cx) = -\ln c - \ln x + \ln 2 - \gamma$ and $\lim_{x \rightarrow 0} K_1(cx) = 1/(cx)$ are used. In addition to the solid lines without symbols, obtained by Eq. (2.13), Fig. 6 shows the two limiting results evaluated by Eqs. (2.16) and (2.17) with dots and triangles, respectively. In general, the result at the origin required in Eq. (2.17) should be infinite, which is artificially set by the value at the origin evaluated from Eq. (2.13) with $Pe_1 = 0.005$ for comparison in Fig. 6. One can see that the result with $Pe_1 = 0.005$ has almost the same shape as that with $Pe_1 = 0$, and the result with $Pe_1 = 10$ is very close to that with $Pe_1 = \infty$.

2.2.3.2 Subsurface temperature rise. Inside the half-plane, the integration of Eq. (2.8) with Eq. (2.5) is not analytically available. Thus the influence coefficient for the *moving* half-plane can be expressed only in an integral form and numerically evaluated. But for the *stationary* half-plane, it is found that

$$g(x_1, x_2) = \int \ln R dx_1 = x_1 \ln R - x_1 + x_2 \tan^{-1}(x_1/x_2), \quad R = \sqrt{x_1^2 + x_2^2} \quad (2.19)$$

$$D^{SS}(x_1, x_2) - D^{SS}(0, x_2) = [g(x_1 - \Delta_1, x_2) - g(x_1 + \Delta_1, x_2) + 2g(\Delta_1, x_2)]/\pi \quad (2.20)$$

It is easy to verify that Eq. (2.17) is a special case of Eq. (2.20) with $x_2 = 0$.

2.3 Frequency Response Functions. Noticing that the x_2 coordinate points inside the half-plane in Fig. 1(b), one can obtain the frequency response function for the half-plane from those for the half-space by setting $\omega_2 = 0$ and changing x_3 into x_2 . Thus,

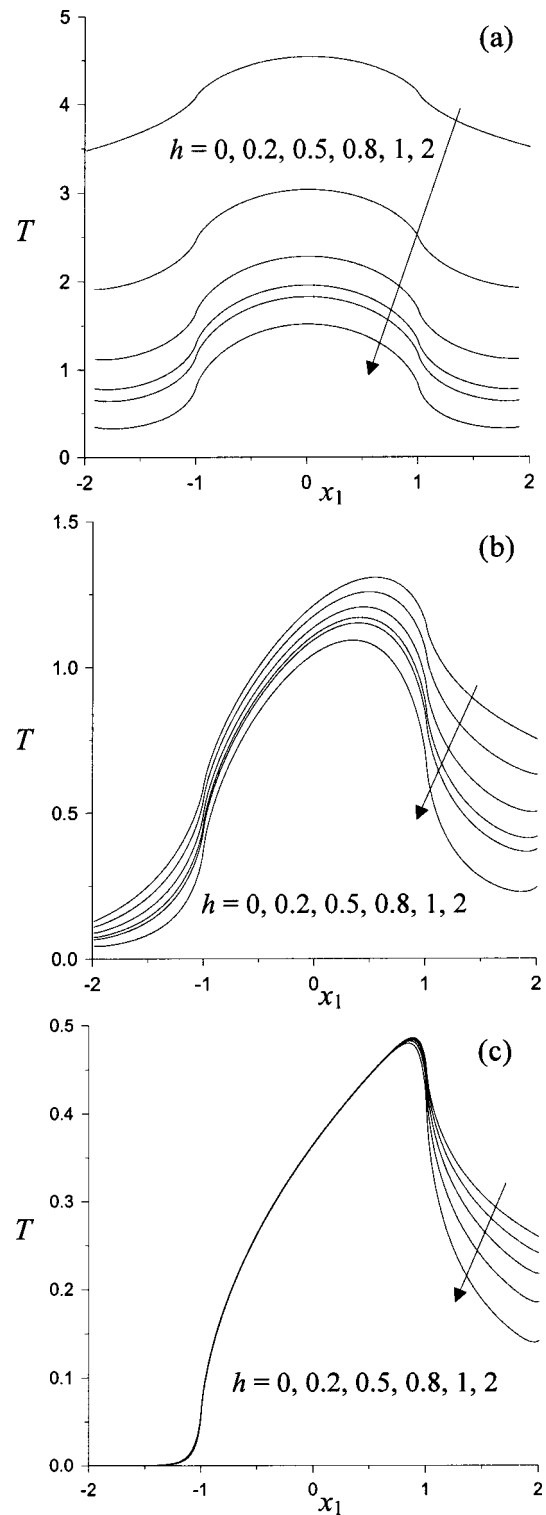


Fig. 8 Surface temperature with different h (half-plane) a) $Pe_1 = 0.005$; b) $Pe_1 = 1$; c) $Pe_1 = 10$

with the new notation, the radius in the frequency domain is defined as $w^2 = \sqrt{\omega_1^2 + i\omega_1 Pe_1}$. For completeness, the frequency response function for each case is listed below,

$$\frac{\tilde{T}(\omega_1, x_2, \omega_t)}{\tilde{q}(\omega_1, \omega_t)} = \frac{\exp(-x_2 \sqrt{w^2 + i\omega_t})}{\sqrt{w^2 + i\omega_t}} \quad (\text{Instantaneous}) \quad (2.21a)$$

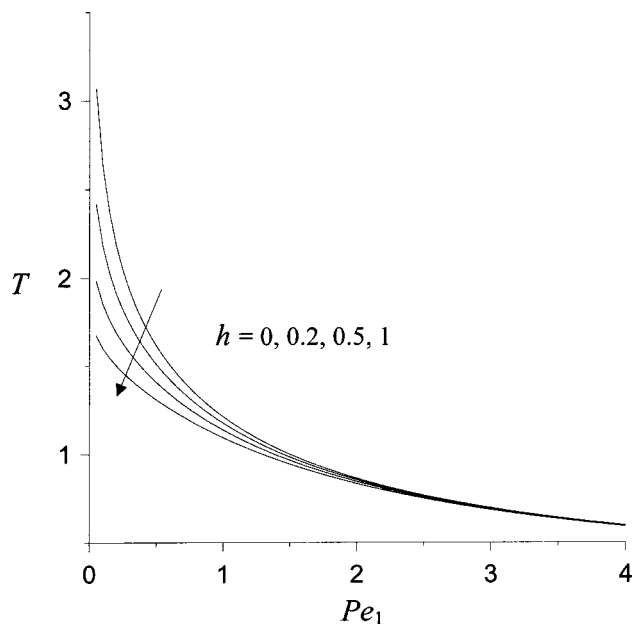


Fig. 9 Variation of temperature at the origin with Péclet number and h (half-plane)

$$\frac{\tilde{T}(\omega_1, x_2, t)}{\tilde{q}(\omega_1)} = \frac{1}{2w} \left[\exp(-x_2 w) \operatorname{erfc}\left(\frac{x_2}{2\sqrt{t}} - w\sqrt{t}\right) - \exp(x_2 w) \operatorname{erfc}\left(\frac{x_2}{2\sqrt{t}} + w\sqrt{t}\right) \right] \quad (\text{Continuous}) \quad (2.21b)$$

$$\frac{\tilde{T}(\omega_1, x_2)}{\tilde{q}(\omega_1)} = \frac{\exp(-x_2 w)}{w} \quad (\text{Steady state}) \quad (2.21c)$$

3 Effect of the Convection Boundary Condition

In previous sections, the adiabatic boundary is assumed outside the surface-heated region S . The effect of the convection boundary condition on the temperature distribution due to surface heating is of interest, since the adiabatic boundary condition may not be reasonable for certain applications. Using the half-space as an example, the convective boundary condition is specified as follows ([5] p. 19):

$$T_{,3} = hT \quad \text{for } (x_1, x_2) \in S \quad (3.1)$$

where h is the dimensionless surface heat-transfer coefficient ($h = hl/K$), which is assumed to be constant.

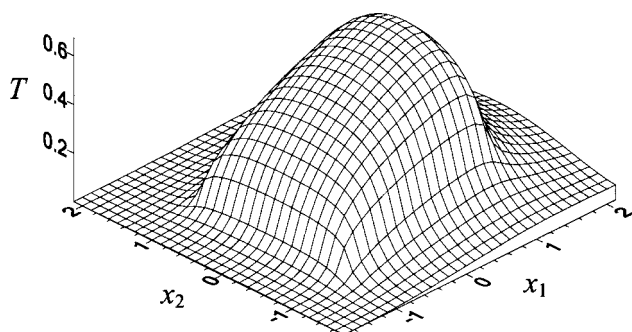


Fig. 10 Surface temperature distribution with $Pe_1=4$ and $h=1$ (half-space)

In this section, the heat source q is assumed to be uniform over $x_i \in [-1, 1]$, $i=1$ or 2 . From the point of view of numerical simulation, complicated heat sources can be solved by the same code as that used for the uniform one. Only the SS surface temperature is considered since it contains the maximum temperature. Fischer et al. [13] found an analytical solution for large Péclet numbers. By introducing a general heat source,

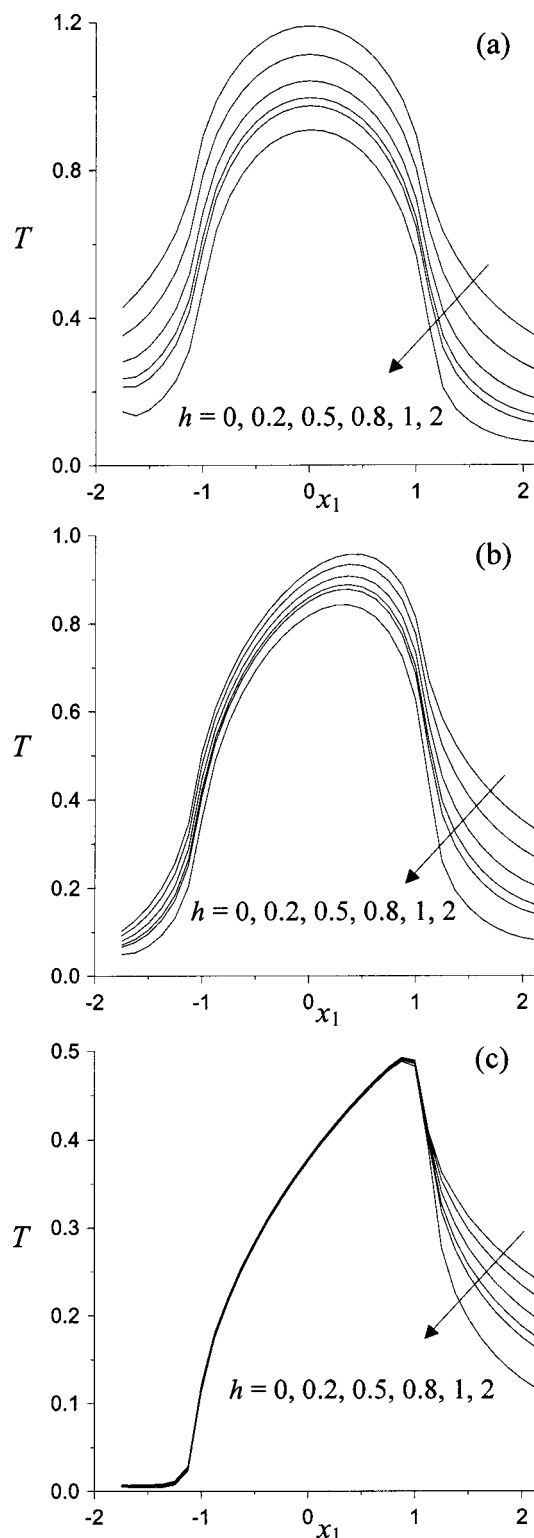


Fig. 11 Surface temperature in x_1 direction with $x_2=0$ and different h (half-space) a) $Pe_1=0$; b) $Pe_1=1$; c) $Pe_1=10$

$$q^*(x_1, x_2) = \begin{cases} q, & (x_1, x_2) \in S \\ -hT, & (x_1, x_2) \notin S \end{cases} \quad (3.2)$$

those formulations with q replaced by q^* presented in Sections 1 and 2 are readily applicable to the problems here. Therefore, the SS temperature rise can be expressed by the following integral:

$$T(x_1, x_2, x_3) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} G^{SS}(x_1 - x'_1, x_2 - x'_2, x_3) q^*(x'_1, x'_2) dx'_1 dx'_2 \quad (3.3)$$

The integral involves the unknown temperature, because q^* depends on temperature. The grid number N is used for discretization. Define $x_j^i = \Delta_j(i - (N/2) - 1)$, $j = 1$ or 2 and $i = 1 \dots N$, and $f(x_1^i, x_2^j, x_3)$ is denoted by $f^{ij}(x_3)$. By using the influence coefficients, one can express the temperature on each grid point as

$$T^{nm} = \sum_{k=1}^N \sum_{j=1}^N (q^*)^{kj} D^{SS}(x_1^n - x_1^k, x_2^m - x_2^j, x_3) \quad (3.4)$$

It should be pointed out that truncation is involved in the rewriting of Eq. (3.3) as Eq. (3.4). This truncation will cause significant error along the border of the computational domain. Equation (3.4) can be rewritten into a linear equation of the temperature field

$$T^{nm} + \sum_{(k,j) \in S} h T^{kj} D^{(n-k)(m-j)} = \sum_{(k,j) \in S} q^{kj} D^{(n-k)(m-j)} \quad (3.5)$$

This equation is in the form of $AT=b$, where A is a matrix $N^2 \times N^2$, and b and T are vectors with N^2 members. This linear equation is solved using the LU decomposition. One should use the analytical expressions of the influence coefficients for the half-plane and the *stationary* or *fast moving* half-space summarized in Sections 1 and 2. If the half-space is *moving* with a moderate Péclet number, the influence coefficients can be obtained numerically from the frequency response functions. An accurate and efficient method to determine the influence coefficients can be found in [14], which uses the fast Fourier technique and is more efficient than the numerical double integration of the Green's function.

3.1 Half-Plane Results. For the half-plane with convection boundary conditions, the linear equation of temperature rise is expressed as follows:

$$T^n + \sum_{k \in S} h T^k D^{(n-k)} = \sum_{k \in S} q D^{(n-k)} \quad (3.6)$$

The temperature distribution along the x_1 axis is shown in Fig. 7 with $h=1$ and different Péclet numbers. By comparing Figs. 6 and 7, one can see that surface convection reduces temperature values significantly with small Péclet number (<1). When the half-plane is moving very slowly, the maximum value is dramatically reduced. Figs. 8(a)–8(c) further show the surface temperature distributions for three specific Péclet numbers (0.005, 1, and 10), respectively. In each plot of Fig. 8, results are shown for six different values of h (0, 0.2, 0.5, 0.8, 1, and 2). For small Péclet numbers [Figs. 8(a) and 8(b)] the whole temperature distribution is substantially reduced by convection. Figure 8(c) shows that when the half-plane is moving from left to right with a large Péclet number, the effect of convection varies based on location. The convection has little effect on the temperature at points in the left-hand side of the plot or inside the surface-heating region, but significantly reduces the temperature at all other points. Figure 9 shows the temperature at the origin as a function of the Péclet number and h . One can conclude that for $Pe_1 > 3$ convection slightly reduces the temperature at the origin, but for small Péclet numbers, convection has a significant effect on the temperature.

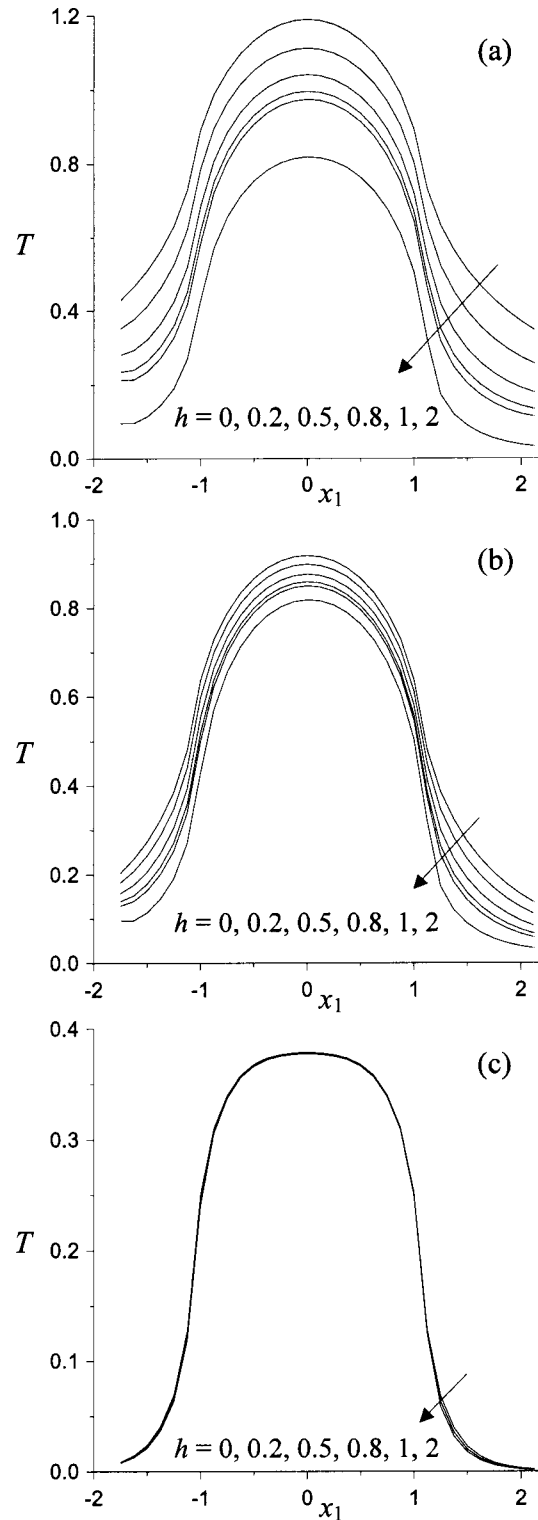


Fig. 12 Surface temperature in x_2 direction with $x_1=0$ and different h (half-space) a) $Pe_1=0$; b) $Pe_1=1$; c) $Pe_1=10$

3.2 Half-Space Results. Since the linear equation of Eq. (3.5) has N^2 unknowns, N is limited by computer memory. The following results are obtained with $N=64$. Figure 10 illustrates the three-dimensional temperature distribution with $Pe_1=4$ and $h=1$, which is typical for the moving half-space. Figures 11(a)–11(c) and 12(a)–12(c) show cross sections in the x_1 directions ($x_2=0$) and in the x_2 directions ($x_1=0$), respectively, with dif-

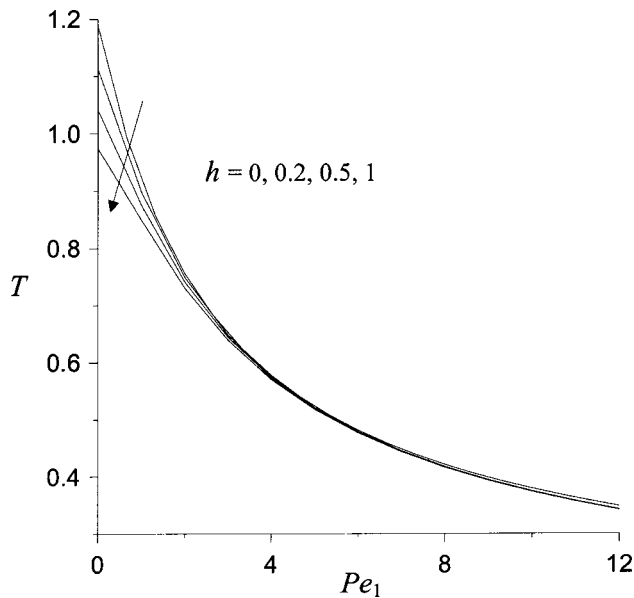


Fig. 13 Variation of temperature at the origin with Péclet number and h (half-space)

ferent values of h under three Péclet numbers (0, 1, and 10). The effect of convection on surface temperature in the half-space is similar to that in the half-plane. For small Péclet numbers the entire computation region is significantly affected by the convection. For large Péclet numbers, the convection hardly affects the temperature rise of points located at $x_1 < 1$ and $\forall x_2$, but significantly affects other points located at $x_1 > 1$. The temperature rise at the origin is obtained for different values of h and shown in Fig. 13 as a function of the Péclet number. It is found that the temperature at the origin is substantially reduced with increasing h when the Péclet number is less than three.

Conclusion

The Green's functions, influence coefficients, and frequency response functions for temperature rise due to surface heating are investigated and summarized. These formulations, which deal with time dependency and component motion, are applicable to half-space and half-plane problems with or without surface convection outside the surface-heated region. The time required to reach approximately steady state is found for moving/stationary half-spaces or stationary half-planes subject to a unit heat flux. It is revealed that a half-plane needs less time than a half-space to achieve a steady state with a reference point located one unit away from the origin. Results also show that with a Péclet number less than three, the convection reduces surface temperature everywhere, while with larger Péclet numbers, the convection significantly reduced the temperature only on the leading side of the surface-heating region.

Nomenclature

$Ei(x)$ = exponential integral
 $\text{erf}(x)$ = error function, $(2/\sqrt{\pi}) \int_0^x \exp(-\tau^2) d\tau$
 $\text{erfc}(x)$ = complementary error function,
 $(2/\sqrt{\pi}) \int_x^\infty \exp(-\tau^2) d\tau$

$\bar{h}; h$ = surface heat-transfer coefficient,
 $J/(\text{m}^2 \text{ K s})$; $h = \bar{h}l/K$
 i = pure imaginary, $\sqrt{-1}$
 K = conductivity, $J/(\text{m K s})$ or $\text{W}/(\text{m K})$
 $K_0(x)$ = modified Bessel function of the second kind of order zero
 $K_1(x)$ = modified Bessel function of the second kind of order one
 l = characteristic length, m
 Pe_j = Péclet number, $\bar{V}_j l / \kappa$
 $\bar{q}; q$ = heat source, W/m^2 (half-space) and W/m (half-plane); $q = \bar{q}/q_0$
 $R = \sqrt{x_1^2 + x_2^2 + x_3^2}$ (half-space) and $R = \sqrt{x_1^2 + x_2^2}$ (half-plane).
 S = surface-heating region
 $\bar{T}; T$ = temperature rise, K; $T = \bar{T}K/(lq_0)$
 $t; t$ = time, s; $t = \kappa \tau / l^2$
 $\bar{V}_j$ = velocity in the x_j direction, m/s
 $\bar{x}_j; x_j$ = coordinate, m; $x_j = \bar{x}_j / l$
 γ = Euler's constant
 κ = thermal diffusivity, m^2/s
 ω_1, ω_2 = frequency domain counterparts of x_1, x_2 , respectively
 ω_t = counterpart of time in the frequency domain
 $_{,j}$ = subscript, partial derivative with respect to x_j coordinate
 $\sim$ = Fourier Transform, e.g., with respect to x_1 ,
 $\int_{-\infty}^{\infty} f(x_1) e^{-i\omega_1 x_1} dx_1$

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